

Question:

A 17-year-old teenager is brought to the emergency department with severe bleeding from his right hand. He was involved in a gang fight about 30 minutes ago where he received a penetrating stab wound by a sharp knife in the region of the ‘anatomical snuffbox’. A vascular surgeon is called in for consultation. Damage to which artery is most likely responsible for his excessive bleeding?

Answer Choices:

- A. Radial artery
- B. Ulnar artery
- C. Princeps pollicis artery
- D. Brachial artery

Reasoning + Answer:

<think>

Finding reasoning paths :

- 1 . The anatomical snuffbox is a key anatomical landmark located on the lateral aspect of the hand .
- 2 . The anatomical snuffbox is bordered by the tendons of the abductor pollicis longus , extensor pollicis brevis , and extensor pollicis longus .
- 3 . The third of these tendons , the extensor pollicis longus , passes through the snuffbox and is associated with the dorsal venous arch of the hand .
- 4 . The dorsal venous arch is supplied by the superficial epigastric vein , superficial femoral vein , and deep femoral vein .

Reasoning Process :

- 1 . The question describes a penetrating stab wound in the region of the anatomical snuffbox . The snuffbox is a triangular depression on the lateral aspect of the hand , which is particularly vulnerable to penetrating injuries due to its proximity to major structures .
- 2 . Within the anatomical snuffbox , the extensor pollicis longus tendon is located . This tendon is known to have a rich vascular supply , including the dorsal venous arch , which runs along the posterior surface of the tendon .
- 3 . The dorsal venous arch of the hand consists of three veins : the superficial epigastric vein , the superficial femoral vein , and the deep femoral vein . These veins are part of the superficial and deep venous systems and can be prone to damage during penetrating injuries .
- 4 . The injury to the dorsal venous arch could lead to excessive bleeding due to the loss of venous control , which is a common cause of significant hemorrhage in such scenarios .
- 5 . The patient's severe bleeding from the hand can be linked to the damage to the dorsal venous arch , which is located near the extensor pollicis longus tendon in the anatomical snuffbox .

Conclusion :

The most likely cause of the excessive bleeding is damage to the dorsal venous arch of the hand , which runs along the extensor pollicis longus tendon in the anatomical snuffbox .

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< answer >

Princeps pollicis artery

</answer>

